Diagnostic Yield and Complications of Transbronchial Lung Cryobiopsy for Interstitial Lung Disease

A Systematic Review and Metaanalysis

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Abstract

Rationale: The diagnostic usefulness and safety of transbronchial lung cryobiopsy for the evaluation of interstitial lung disease remain unclear.

Objectives: This systematic review and metaanalysis aims to establish the diagnostic accuracy and yield of transbronchial cryobiopsy for interstitial lung diseases.

Methods: We searched MedLine, EMBASE, Cochrane Central Register of Controlled Trials, and conference proceedings to identify studies assessing the diagnostic accuracy (compared with surgical biopsy) or yield of transbronchial lung cryobiopsy for interstitial lung disease (from database inception to January 2016). The diagnostic accuracy and yield were quantified and stratified by the method of diagnosis determination (histologic interpretation in isolation vs. incorporation within a multidisciplinary discussion). The frequency of procedure-related complications was also assessed from these reports. For full-text studies, random-effects models were used to calculate pooled estimates of diagnostic accuracy, yield, and complication frequency.

Measurements and Main Results: Of 900 citations, 11 studies were selected for inclusion in this systematic review (7 full text, 4 abstracts). The selected studies reported on a total of 731 patients. No studies reported the diagnostic accuracy of transbronchial cryobiopsy. Diagnostic yield ranged from 74 to 98% when transbronchial cryobiopsy findings were interpreted in isolation, with a pooled estimate of 83% (95% confidence interval [CI], 73–94). Diagnostic yield ranged from 51 to 98% when transbronchial cryobiopsy was reviewed within a multidisciplinary discussion, with a pooled estimate of 79% (95% CI, 65–93). Pooled estimates for pneumothorax and moderate/severe bleeding were 12% (95% CI, 3–21) and 39% (95% CI, 3–76), respectively.

Conclusions: The diagnostic accuracy of transbronchial lung cryobiopsy cannot be determined given the absence of studies directly comparing cryobiopsy diagnoses with diagnoses derived from surgical lung biopsies interpreted within multidisciplinary discussions. The histopathological and multidisciplinary discussion-based diagnostic yield of transbronchial cryobiopsy appears high, but with variable frequencies of complications dominated by pneumothorax and moderate-to-severe hemorrhage.

Keywords: diagnosis; diffuse parenchymal lung diseases; pulmonary fibrosis

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Interstitial lung diseases (ILDs) are a large and heterogeneous group of disorders characterized by inflammation and/or fibrosis of the pulmonary interstitium. ILD is challenging to diagnose, because many subtypes have overlapping clinical and radiological features (1, 2). The diagnostic gold standard for ILD is multidisciplinary discussion and the real-time integration of clinical, radiological, and pathological data by specialists with expertise in ILD (1–3). A diagnosis can be established with a high degree of confidence in the absence of histopathology in some patients with ILD (1, 4), but a substantial minority are considered unclassifiable, leaving the patient without a confident diagnosis or a clear management plan (5).

Transbronchial lung cryobiopsy has been proposed as an alternative bronchoscopic technique for histological sampling in patients with ILD, potentially combining the higher yield of surgical lung biopsy with the lower complication rate of transbronchial forceps biopsy (6, 7). Transbronchial cryobiopsy has gained support rapidly from some groups as a viable technique in the evaluation of ILD, but there is significant heterogeneity in the existing literature. Three recent systematic reviews and metaanalyses have concluded that transbronchial cryobiopsy provides a high diagnostic yield with acceptably low complication rates (8–10). However, these reports are limited by an inability to assess diagnostic accuracy compared with established gold standards, as well as by inclusion of non-ILD populations, including patients undergoing assessment for lung masses, those undergoing post–lung transplant rejection surveillance, and immunocompromised patients with diffuse pulmonary disease.

The heterogeneity of individual studies and the limitations of previous systematic reviews highlight the need for a more comprehensive and rigorous assessment of the existing evidence. We therefore performed a systematic review and metaanalysis of the literature to identify studies evaluating the histopathologic and/or multidisciplinary discussion–based diagnostic accuracy and yield of transbronchial cryobiopsy in the evaluation of patients with ILD. The frequencies of procedure-related complications associated with transbronchial cryobiopsy were also identified from included studies.

Methods

Data Sources and Searches

We performed a systematic review and metaanalysis after a predetermined protocol in accordance with the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines (11). Two reviewers (K.A.J. and V.S.M.) independently identified potentially relevant articles in any language by searching Medline, EMBASE, and the Cochrane Central Register of Controlled Trials (from database inception to January 2016). Searches were designed with the goal of identifying all publications that reported cryobiopsy in patients with ILD. For electronic database searches, three search themes were specified:

1. To identify terms related to the procedure of interest, the first Boolean search used the term "or" to explode (search by subject heading) and map (search by keyword) the medical subject heading (MeSH) heading "cryo*."
2. To identify terms related to the disease process of interest, the second Boolean search used the term "or" to explode and map the MeSH headings "interstitial" or "parenchyma*" or "fibro*" or "diffuse."
3. To identify terms related to the organ system of interest, the third Boolean search used the term "or" to explode and map the MeSH headings "pulm*" or "lung."

The three Boolean search themes were combined using the Boolean term "and." The search was not limited to specific study designs and did not include the terms "diagnosis" or "complication," to maximize the number of returned studies. Additional potential articles and unpublished studies were identified from online repositories and conference proceedings (up to March 2016). Local institutional review board approval was not required for this study.

Study Selection

Two reviewers independently screened all citations for eligibility using sequential review of titles, abstracts, and full publications. Studies of any design were included if they met the prespecified criteria of including at least 10 consecutively enrolled patients with suspected ILD and reported diagnostic accuracy or yield only for patients with ILD on the basis of either histopathologic findings or via incorporation of histopathology in a multidisciplinary discussion. Citations were excluded if they did not present original data, and disagreements were resolved through consensus. We included the study with the largest sample size to prevent double counting of patients in the event of multiple publications with overlapping recruitment periods at a single center.

Data Extraction and Quality Assessment

Two reviewers independently extracted data from included studies, with disagreements resolved via consensus. The coprimary outcomes were diagnostic accuracy and yield. Diagnostic accuracy was defined as the establishment of a diagnosis consistent with that obtained via the current gold standard of surgically obtained lung tissue reviewed in a multidisciplinary discussion (2), whereas diagnostic yield was defined as the achievement of a satisfactory ("confident," "definite," or "probable") clinical diagnosis without comparison with the current gold standard. Both outcomes were stratified by whether these were based on the interpretation of transbronchial cryobiopsy findings in isolation or by incorporation within a multidisciplinary discussion.

Prespecified safety outcomes included procedure-related complications including bleeding, pneumothorax, acute exacerbation of ILD, and death. Bleeding complications and severity were reclassified where possible on the basis of details in the original publication, according to British Thoracic Society guidelines (12), with moderate and severe bleeding events pooled in the metaanalysis. Bleeding was classified as moderate if adequate bleeding control required intubation of the biopsied segment with the bronchoscope in the wedge position or the use of cold saline or adrenaline. Bleeding was considered severe if it necessitated placement of a bronchus blocker or sealant; required resuscitation, blood transfusion, or critical care unit admission; or resulted in death.

Additional data extracted included study design, sample size, population demographics (mean age, sex distribution), procedural technique (location of procedure, mode of sedation, use of rigid bronchoscope, fluoroscopic guidance, and endotracheal intubation), mean/median biopsy size, and adequacy of biopsy per case or per biopsy (defined by the presence of alveolated tissue). Two reviewers independently assessed study quality using a component approach, with specific items adapted from the Downs and Black Scale (13).

Statistical Methods and Metaanalysis

Descriptive statistics, including stratification by method of diagnosis determination (interpretation of transbronchial cryobiopsy findings in isolation vs. incorporation within a multidisciplinary discussion), were used to quantify the diagnostic accuracy and yield. For full-text studies, DerSimonian and Laird random-effects models were used to calculate pooled estimates of diagnostic yield and complications (14). These were reported as proportions with corresponding 95% confidence intervals (CIs).

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Heterogeneity among studies was assessed using visual inspection of forest plots and the calculation of the Cochran Q and I^2 statistics, the latter quantifying the percentage of variation attributable to between-study differences (15). The frequency of procedure-related complications was assessed similarly. All analyses were performed using Stata 12.1 (StataCorp, College Station, TX).

Results

Search Results

The initial search identified 900 citations, and 611 abstracts were reviewed after removal of duplicates (Figure 1). Of these, 30 publications underwent full-text review, and 11 studies (7 full text, 4 conference abstracts) with unique populations were included in the final systematic review.

Study Characteristics

The 11 eligible studies included 731 patients (sample size range, 24–297), with 553 patients from the seven full-text publications (Table 1) (8, 16–25). There were six retrospective cohorts, two prospective cohorts, one randomized controlled trial, and two studies with an unclear design. Studies originated from six countries and were published between 2013 and 2016. Seven studies reported mean age, which ranged from 56.2 to 66.2 years, and nine studies reported sex distribution, ranging from 33 to 69% male. Mean FVC and diffusion capacity of the lung for carbon monoxide were reported in five studies, ranging from 69 to 86% predicted and 50 to 68% predicted, respectively. The mean biopsy area was reported in eight studies, ranging from 6.6 to 64.2 mm^2. All studies reported diagnostic yield and procedure-related complications, but none reported diagnostic accuracy compared with established gold standards.

Study Quality

The quality of full-text studies included in the metaanalysis was variable (Table 2). The majority were of good quality, although only five of seven (71%) clearly described the study population and six of seven (86%) reported all prespecified adverse events. External validity was low because only four of seven (57%) clearly reported whether participants were representative of the entire population from which they were recruited.

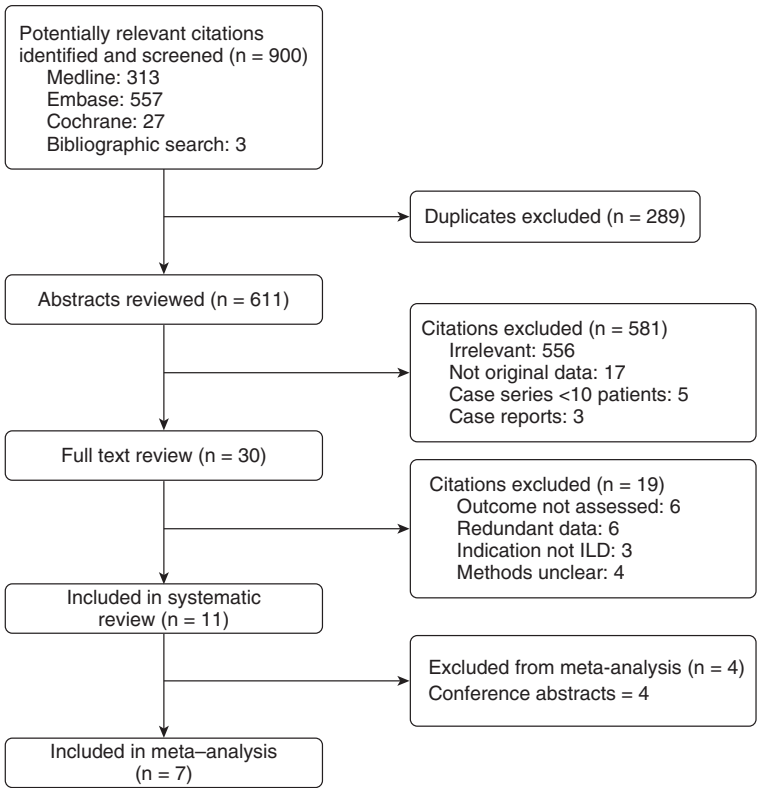

Figure 1. Details of study selection for review. ILD = interstitial lung disease.

Diagnostic Accuracy and Yield

The diagnostic accuracy of transbronchial cryobiopsy could not be determined because no studies directly compared cryobiopsy with the gold standard of surgical lung biopsy incorporated into a multidisciplinary discussion. Seven studies reported diagnostic yield from histopathologic assessment alone (mean, 78%; range, 74–98%), with a pooled diagnostic yield of 83% (95% CI, 73–94) in the five full-text publications reporting this outcome (Table 3 and Figure 2). Eight studies reported diagnostic yield after cryobiopsy tissue was incorporated into a multidisciplinary discussion (mean, 86%; range, 51–98%), with a pooled diagnostic yield of 79% (95% CI, 65–93) in the six full-text publications reporting this outcome (Table 3 and Figure 3). Visual assessment of the forest plots demonstrated significant heterogeneity in the yield for both histopathologic assessment alone and incorporation into a multidisciplinary discussion ($P < 0.001$, $I^2 > 90\%$). An additional study was identified that compared the impact of transbronchial cryobiopsy with that of surgical biopsy on the diagnostic process in the context of a multidisciplinary discussion; however, this study did not report the yield of cryobiopsy and was therefore excluded from this analysis (26).

Of the full-text publications, four included a subset of patients who underwent surgical lung biopsy after a nondiagnostic transbronchial cryobiopsy, one prospectively compared cryobiopsy with transbronchial biopsy, one retrospectively compared transbronchial cryobiopsy with surgical lung biopsy, and one did not compare cryobiopsy with other biopsy methods. A total of 13 patients from four studies underwent surgical lung biopsy after a nondiagnostic transbronchial lung cryobiopsy. The final multidisciplinary discussion–based diagnosis was idiopathic pulmonary fibrosis in seven patients (54%), cryptogenic organizing pneumonia in two patients (15%), and hypersensitivity pneumonitis, nonspecific interstitial pneumonia, peribronchial metaplasia,

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Table 1. Study characteristics

Study	Study Design	Country	No. Patients	Age, Mean (Range) or Median (SD) (*yr*)	Male (%)	FVC % Predicted (Range or SD)	D_{LCO} % Predicted (Range or SD)
Full-text publications							
Fruchter 2014 (18)	Retrospective cohort study	Israel	75	56.2 (17–81)	54.7	Not reported	Not reported
Griff 2014 (19)	Retrospective cohort study	Germany	52	63 (13)	69	Not reported	Not reported
Hagmeyer 2015 (20)	Retrospective cohort study	Germany	32	65.4 (45–83)	68.8	74.6 (40–123)	52.4 (17–108)
Hernández-González 2015 (21)	Retrospective cohort study	Spain	33	64 (30–79)	33	69 (43–103)	50 (23–82)
Kropski 2013 (22)	Retrospective cohort study	United States	25	57.1 (27–75)	52	75.3 (17.0)	66.2 (25.3)
Pajares 2014 (25)	Randomized controlled trial	Spain	39	60.3 (10.3)	51.3	78.2 (15.2)	67.5 (19.8)
Ravaglia 2016 (8)	Prospective cohort study	Italy	297	60 (21–78)	57.9	86 (37–137)	58.8 (14–121)
Conference abstracts							
Echevarria-Uraga 2015 (16)	Unclear	Spain	85	Not reported	Not reported	Not reported	Not reported
Elshafi 2015 (17)	Prospective	Ireland	24	66.2	66	Not reported	Not reported
Martin 2014 (23)	Unclear	Spain	36	59.1	55.5	Not reported	Not reported
Oberle 2014 (24)	Prospective	Germany	33	Not reported	Not reported	Not reported	Not reported

Definition of abbreviation: D_{LCO} = diffusion capacity of the lung for carbon monoxide; FVC = forced vital capacity; SD = standard deviation.

and unclassifiable in one patient each (18, 20–22). One prospective randomized controlled trial reported higher diagnostic yields with transbronchial cryobiopsy compared with conventional transbronchial biopsy (74.4% vs. 34.1% for histopathologic findings interpreted in isolation and 51.4% vs. 29.1% for findings incorporated within a multidisciplinary discussion) (25).

Procedure-related Complications

Complications were reported in all studies with pneumothorax frequencies ranging from 0 to 25.9% (mean, 8.8%). Of the five studies reporting nonzero values, the pooled frequency of procedure-related pneumothorax was 12% (95% CI, 3–21) (Figure 4). There was significant heterogeneity in the frequency of pneumothorax ($P < 0.001$, $I^2 = 89.3\%$). Ten studies reported the frequency of moderate/severe bleeding (mean, 26.6%; range, 0–78%). Of the four studies reporting nonzero values, the pooled frequency of procedure-related moderate/severe bleeding was 39% (95% CI, 3–76) (Figure 5). There was significant heterogeneity in the frequency of moderate/severe bleeding ($P < 0.001$, $I^2 = 97.6\%$). Pooled analyses of acute exacerbation and procedure-related death were not performed because there was only one of each event in all studies combined.

The only prospective randomized trial comparing transbronchial cryobiopsy with transbronchial biopsy reported similar frequencies of procedure-related complications (25). Pneumothorax occurred in 7.7% of cryobiopsy vs. 5.2% of transbronchial biopsy ($P = 0.99$), and moderate bleeding occurred in 56.4% of transbronchial cryobiopsy vs. 34.2% of transbronchial biopsy ($P = 0.07$). Cryobiopsy was associated with fewer days of hospitalization than surgical lung

Table 2. Study quality

Study	Reporting						External Validity
	Is the Aim/Objective of the Study Clearly Described?	Are the Main Outcomes To Be Measured Clearly Described in the Introduction or Methods?	Are the Characteristics of the Patients Included in the Study Clearly Described?	Is the Procedure of Interest Clearly Described?	Are the Main Findings of the Study Clearly Described?	Have All Important Adverse Events that May Be a Consequence of the Intervention Been Reported?	Were the Subjects Asked To Participate in the Study Representative of the Entire Population from which They Were Recruited?
Fruchter 2014 (18)	Yes	Yes	No	Yes	Yes	Yes	Yes
Griff 2014 (19)	Yes	Yes	No	Yes	Yes	No	Unclear
Hagmeyer 2015 (20)	Yes	No	Yes	Yes	No	Yes	Yes
Hernández-González 2015 (21)	Yes	Yes	Yes	Yes	Yes	Yes	Unclear
Kropski 2013 (22)	Yes	Yes	Yes	Yes	Yes	Yes	Unclear
Pajares 2014 (25)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ravaglia 2016 (8)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

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Table 3. Main study outcomes

Study	Method of Diagnosis	Histology Diagnoses [No. per Patient unless Otherwise Stated (%)]	Histology Yield (%)	No. MDD Diagnoses (%)	MDD Yield (%)	Complication Rates (%)
Full-text publications						
Fruchter 2014 (18)	Histology and MDD	22 Interstitial fibrosis (29.3) 21 NSIP (28) 11 COP (14.7) 7 UIP (9.3) 3 PLCH (4) 2 Normal lung tissue (2.7) 1 LAM (1.3) 1 Sarcoidosis (1.3) 1 Lipoid pneumonia (1.3) 1 Alveolar proteinosis (1.3) 1 DIP (1.3) 1 Hypersensitivity pneumonitis (1.3) 1 Silicosis (1.3) 1 Eosinophilic pneumonia (1.3) 1 Lymphangitic carcinomatosis (1.3)	98	52 Definite ILD diagnosis (70) 21 Probable ILD diagnosis (28) 2 Nondiagnostic (2.7)	98	2 Pneumothorax (2.6) 3 Moderate bleed (4)
Griff 2014 (19)	Histology and MDD	10 Sarcoidosis (19.2) 9 IPF (17.3) 8 COP (15.4) 6 HP (11.5) 2 Rheumatoid lung disease (3.8) 2 Medically induced lung damage (3.8) 1 Alveolar microlithiasis (1.9) 1 NSIP (1.9) 1 Scleroderma (1.9) 1 Histiocytosis (1.9) 1 p-ANCA-pos vasculitis	79	13 IPF 12 Sarcoidosis 9 COP 7 HP 2 Rheumatoid lung disease 2 Medically induced lung damage 2 Scleroderma 1 Alveolar microlithiasis 1 NSIP 1 pANCA-pos vasculitis	79	0 Severe bleeding (0) Moderate bleeding not reported
Hagmeyer 2015 (20)	MDD	Not reported	Not reported	Not reported	72	6 Pneumothorax (19) 17 Severe bleed (53) 8 Moderate bleed (25)
Hernández-González 2015 (21)	Histology and MDD	19 Inconsistent with UIP (57.6) 5 Probable UIP (15.2) 5 Nondiagnostic (15.2) 2 UIP (6.1) 2 Invalid (6.1)	79	Probable histologic UIP: 3 IPF (9.1) 1 UIP autoimmune (3) 1 Drug-induced (3) Histologic UIP: 1 IPF (3) 1 UIP autoimmune (3) Of those Inconsistent with UIP: 6 HP (18.2) 5 Idiopathic NSIP (15.1) 3 Systemic autoimmune disease associated NSIP (9.1) 1 LIP (3) 1 Sarcoidosis (3) 1 LAM (3) 1 COP (3) 1 Peribronchiolar metaplasia (3) Invalid histology: 2 ILD of unknown origin (6.1) Nondiagnostic: 2 Idiopathic NSIP (6.1) 2 ILD of unknown origin (6.1) 1 Peribronchiolar metaplasia (3)	88	4 Pneumothorax (12) 7 Moderate bleed (21)
Kropski 2013 (22)	Histology and MDD	Not reported	76	7 UIP/IPF (28) 5 Nondiagnostic (20) 2 BOOP/COP (8) 2 RB-ILD/DIP (8) 2 Drug induced (8) 2 Malignancy (8) 1 HP (4) 1 Constrictive bronchiolitis (4) 1 Bronchiolitis obliterans (4) 1 Normal (4) 1 Nonviable sample (4)	80	1 Post-procedural hypoxemia (4)
Pajares 2014 (25)	Histology and MDD	12 NSIP (30.8) 10 No histologic diagnosis provided (25.6) 7 UIP (17.9) 3 HP (7.7) 3 Organizing pneumonia (7.7) 2 Bronchiolitis-associated diffuse ILD (5.1) 1 Diffuse alveolar damage (2.6) 1 Sarcoidosis (2.6)	74	19 No MDD diagnosis provided (48.7) 10 NSIP (25.6) 3 COP (7.7) 3 HP (7.7) 2 RB-ILD (5.1) 1 Acute alveolar injury (2.6) 1 Sarcoidosis (2.6)	51	3 Pneumothorax (7.7) 22 Moderate bleed (56.4)

(*Continued*)

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Table 3. (*Continued*)

Study	Method of Diagnosis	Histology Diagnoses [No. per Patient unless Otherwise Stated (%)]	Histology Yield (%)	No. MDD Diagnoses (%)	MDD Yield (%)	Complication Rates (%)
Ravaglia 2016 (8)	Histology	92 UIP (31.0) 51 Nondiagnostic pattern (17.2) 36 Other (12.1) (neoplasms, eosinophilic pneumonia, follicular bronchiolitis, alveolar proteinosis, vasculitis, acute fibrinous and organizing pneumonia, DIPNECH or PLCH) 31 Organizing pneumonia (10.4) 25 NSIP (8.4) 24 HP (8.1) 22 Sarcoidosis, other granulomatosis (7.4) 12 DIP/RB-ILD (4.0) 4 Diffuse alveolar damage (1.3)	83	n/a	n/a	60 Pneumothorax (20) 2 Respiratory failure (0.7) 1 Acute exacerbation (0.3) 1 Death (0.3)
Conference abstracts						
Echevarria-Uraga 2015 (16)	Histology	UIP reported as the most common histological pattern (39) Other diagnoses not reported	Not reported	n/a	n/a	1 Pneumothorax (1.2) 10 Moderate bleed (11.8) 2 Respiratory failure (2.4) 1 Acute exacerbation (1.2)
Elshafi 2015 (17)	Histology	UIP (23/77 samples = 29.9) Granulomatous disease (2/77 samples = 2.6) Eosinophilic pneumonia (1/77 samples = 1.3)	96	n/a	n/a	7 Pneumothorax (25.9) 3 Moderate bleed (11.1)
Martin 2014 (23)	MDD	n/a	n/a	Accurate diagnosis: 7 UIP (19.4) 4 HP (11.1) 3 Sarcoidosis (8.3) 1 Drug-induced ILD (2.7) 1 Pulmonary alveolar proteinosis (2.7) 1 Emphysema (2.7) 1 Alveolar hemorrhage (2.7) 1 NSIP due to CTD (2.7) 1 COP (2.7) 1 Adenocarcinoma (2.7) High probability diagnosis: 2 UIP (5.5) 1 HP (2.7) 1 DIP (2.7) 1 Drug-induced vs. acute eosinophilic pneumonia (2.7) 1 COP (2.7) No diagnosis: 7 Unclassifiable (19.4) 2 No lung parenchyma (5.5)	75	5 Pneumothorax (14.3) 3 Moderate bleed (8.3) 1 Severe bleed (2.8)
Oberle 2014 (24)	MDD	n/a	n/a	8 UIP related to CTD (24.2) 7 Nonspecific changes (21.2) 5 HP (15.2) 4 IPF (12.1) 3 Sarcoidosis (9.1) 3 RB-ILD (9.1) 2 NSIP (6.1) 1 Pneumoconiosis (3.0)	79	1 Pneumothorax (3)

Definition of abbreviations: BOOP = bronchiolitis obliterans organizing pneumonia; COP = cryptogenic organizing pneumonia; CTD = connective tissue disease; DIP = desquamative interstitial pneumonia; DIPNECH = diffuse idiopathic pulmonary neuroendocrine cell hyperplasia; HP = hypersensitivity pneumonitis; ILD = interstitial lung disease; IPF = idiopathic pulmonary fibrosis; LAM = lymphangioleiomyomatosis; LIP = lymphocytic interstitial pneumonia; MDD = multidisciplinary discussion; n/a = not available; NSIP = nonspecific interstitial pneumonia; pANCA = perinuclear antineutrophil cytoplasmic antibodies; PLCH = pulmonary Langerhans cell histiocytosis; RB-ILD = respiratory bronchiolitis-interstitial lung disease; UIP = usual interstitial pneumonia.

biopsy (2.6 vs. 6.1; $P < 0.001$) and 1 of 297 patients (0.003%) died in the cryobiopsy group compared with 4 of 250 (0.016%) in the surgical lung biopsy group (8).

Discussion

In this metaanalysis of seven full-text studies comprising 553 patients, we show a pooled diagnostic yield for transbronchial lung cryobiopsy of approximately 80% in patients being evaluated for suspected ILD. However, there is significant heterogeneity across studies that is likely driven by variations in study design, underlying disease processes, and mode of determining the diagnosis. A major limitation of previous studies is that a simple yield is not the most clinically informative outcome measure in the assessment of a new diagnostic modality. To date, no studies have compared the performance of transbronchial cryobiopsy with the gold standard of surgical lung biopsy incorporated into a multidisciplinary discussion. The lack of comparison with an established gold standard prohibits any firm conclusions regarding the usefulness of transbronchial cryobiopsy and indicates that further evidence is needed before

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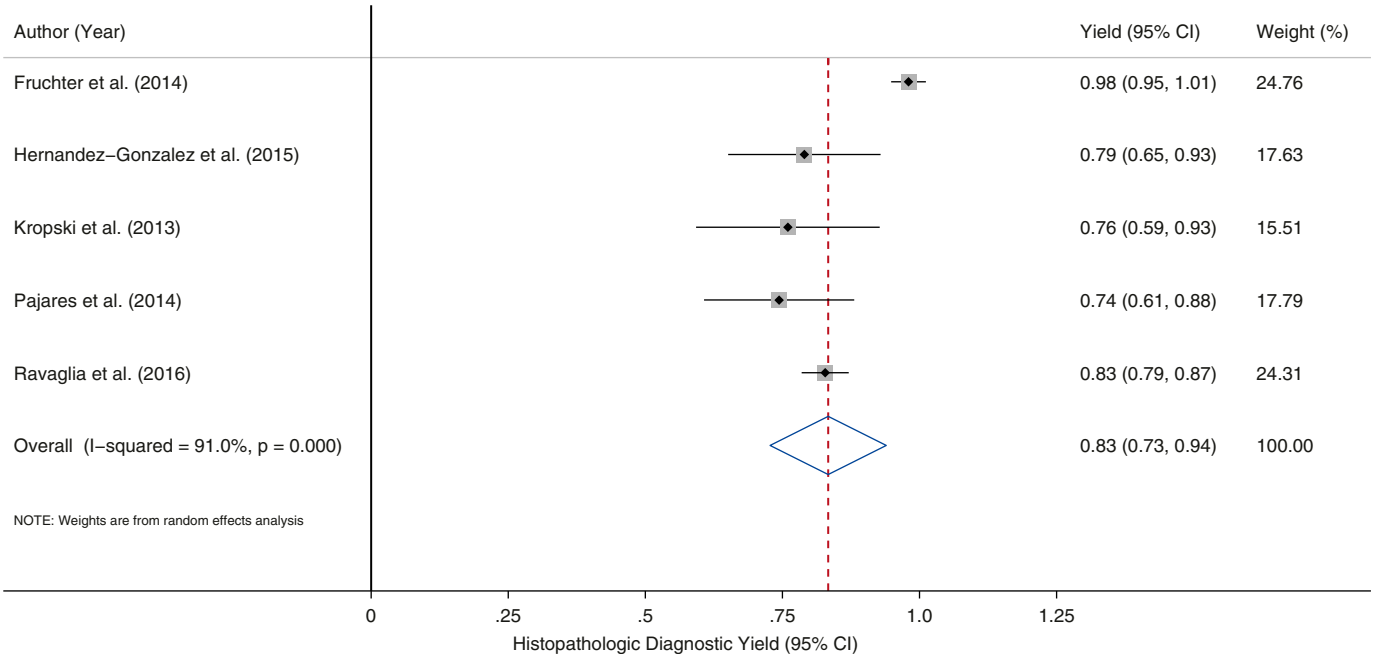

Figure 2. Forest plot of histopathologic diagnostic yield. CI = confidence interval.

endorsing this technique for the evaluation of suspected ILD.

Previous systematic reviews have sought to summarize the usefulness and safety of transbronchial cryobiopsy, but these reviews have limitations that emphasize the need for a more rigorous assessment of the diagnostic usefulness of cryobiopsy. First, none have exclusively included or reported on patients with suspected ILD. The diagnostic yields and complications of cryobiopsy likely differ depending on the underlying disease process, and thus, an assessment of cryobiopsy in an ILD-only cohort is

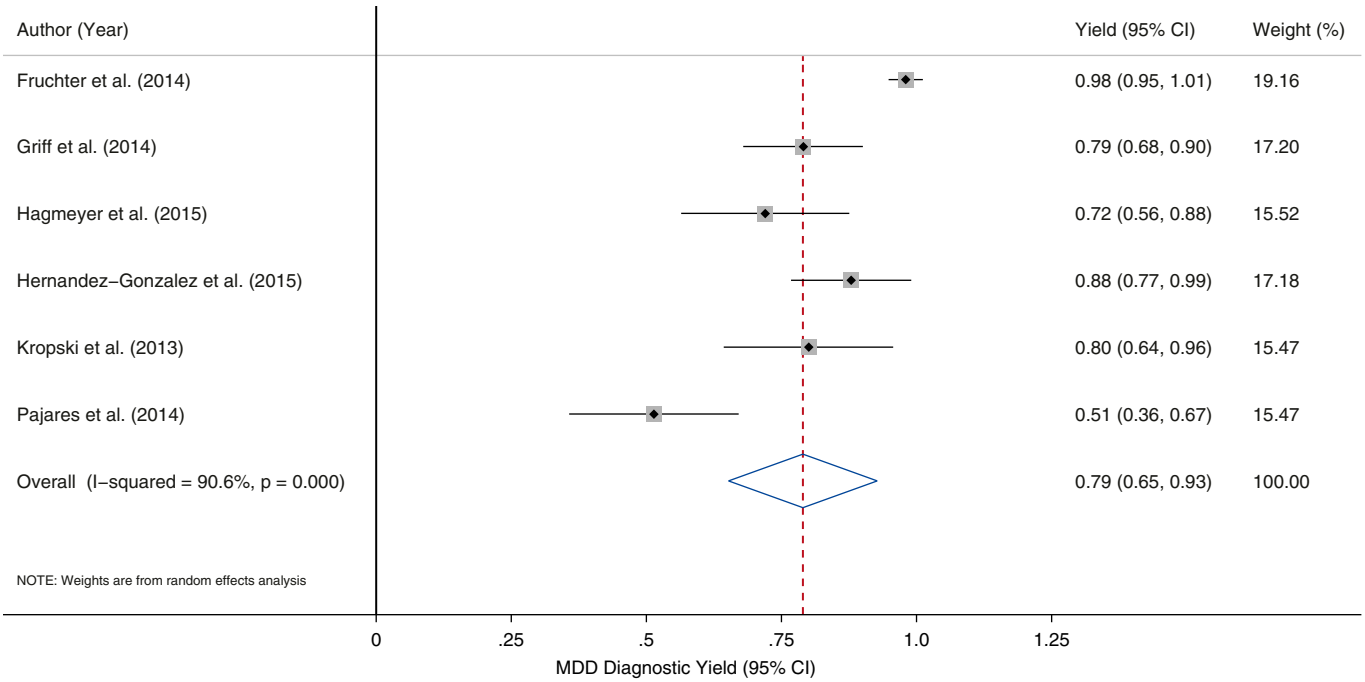

Figure 3. Forest plot of multidisciplinary discussion-based diagnostic yield. CI = confidence interval; MDD = multidisciplinary discussion.

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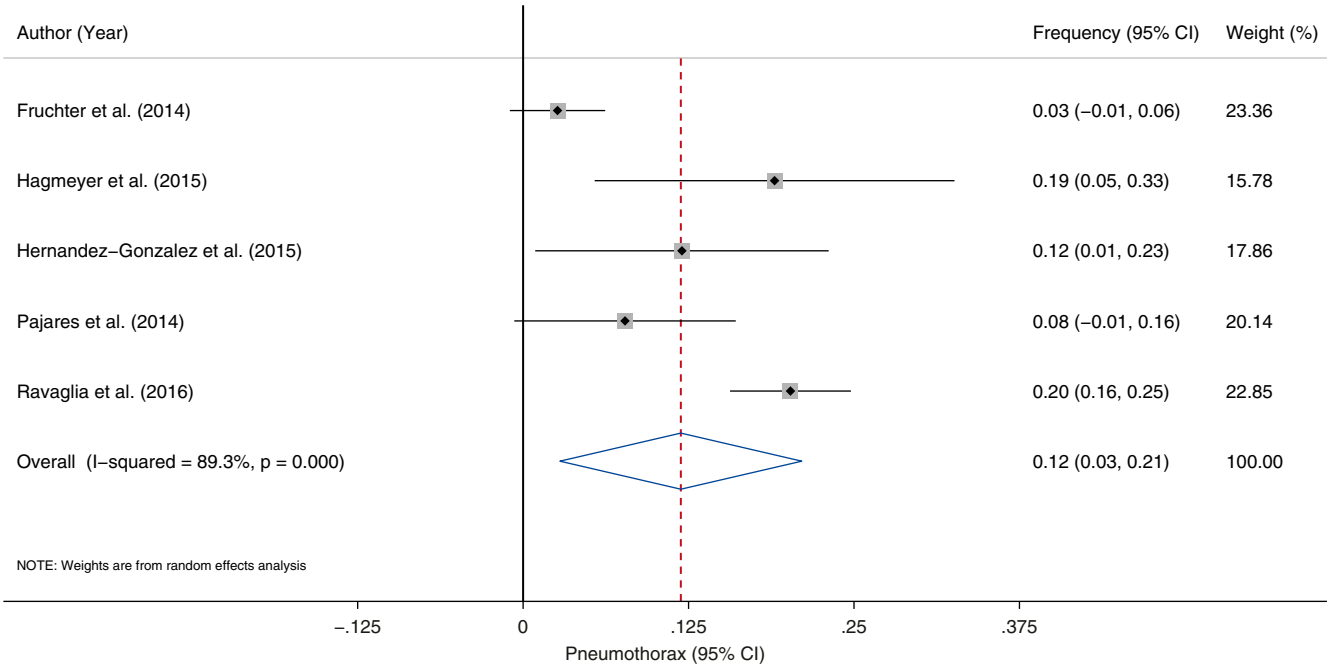

Figure 4. Forest plot of procedure-related pneumothorax. CI = confidence interval.

essential to assess its usefulness in this group. Second, previous systematic reviews have not commented on the importance of comparing transbronchial cryobiopsy results with established gold standards, resulting in potentially optimistic estimates of the overall usefulness of cryobiopsy when presented as a simple yield. As noted previously, we believe diagnostic accuracy is a much more meaningful outcome measure than is diagnostic yield. Finally, there is a rapidly expanding literature base for cryobiopsy, and we have expanded on previous systematic reviews to include the most recently published data on the topic, making this, to the best of our knowledge, the most current assessment of cryobiopsy for suspected ILD.

The wide variety of ILDs in previous studies suggests that transbronchial cryobiopsy can support a confident diagnosis in many clinical scenarios. However, there is inadequate evidence that cryobiopsy reliably distinguishes fibrotic ILD subtypes, a common and challenging clinical scenario. The recommendation to sample at least two involved lobes during a surgical lung biopsy is based on the potential for sampling error and discordant findings if fewer samples are taken (27, 28). Although typically well preserved and free of crush artifact, transbronchial cryobiopsy samples are smaller and are taken from more central regions, compared with surgically obtained samples, and may be less likely to adequately represent the underlying histology. Furthermore, cryobiopsy samples are often taken from a single lobe, and thus, discordant patterns may not be identified. These issues again highlight the need for future studies that directly compare transbronchial cryobiopsy findings with surgical lung biopsy.

A recent study on the multidisciplinary discussion process compared the impact of samples obtained by transbronchial lung cryobiopsy with that of surgically obtained lung tissue (26). This well-performed study was excluded from our metaanalysis because of the absence of a reported diagnostic yield (8). This study included patients with radiographic evidence of fibrotic ILD who underwent either cryobiopsy (n = 58) or surgical lung biopsy (n = 59) as part of their diagnostic evaluation. Through a stepwise multidisciplinary discussion approach, the authors showed that the addition of cryobiopsy changed the initial clinical-radiologic diagnosis less frequently than did surgical lung biopsy (26% vs. 36% of cases) and that interobserver agreement for a usual interstitial pneumonia pattern was lower for patients undergoing the transbronchial cryobiopsy than for those undergoing surgical lung biopsy (kappa values of 0.59 vs. 0.86). These findings, although from a single, nonvalidated study, suggest potential sampling limitations of cryobiopsy compared with surgically obtained tissue. To the best of our knowledge, this is the only study to date to describe the role of transbronchial cryobiopsy in the context of a multidisciplinary discussion; however, these results raise additional questions regarding the diagnostic accuracy of cryobiopsy.

There are important limitations to the current gold standard for the diagnosis of ILD, even when surgical lung biopsy is incorporated into a multidisciplinary discussion. At best, there is moderate-good interobserver agreement by expert pathologists interpreting ILD patterns (29, 30), with potentially wider variation in nonacademic practice. Although the multidisciplinary discussion remains the gold standard for improving diagnostic agreement within a team (3, 31), recent data testing the intermultidisciplinary discussion agreement show good agreement for idiopathic pulmonary fibrosis and connective tissue disease–related ILD, moderate agreement for nonspecific interstitial pneumonia, and fair agreement

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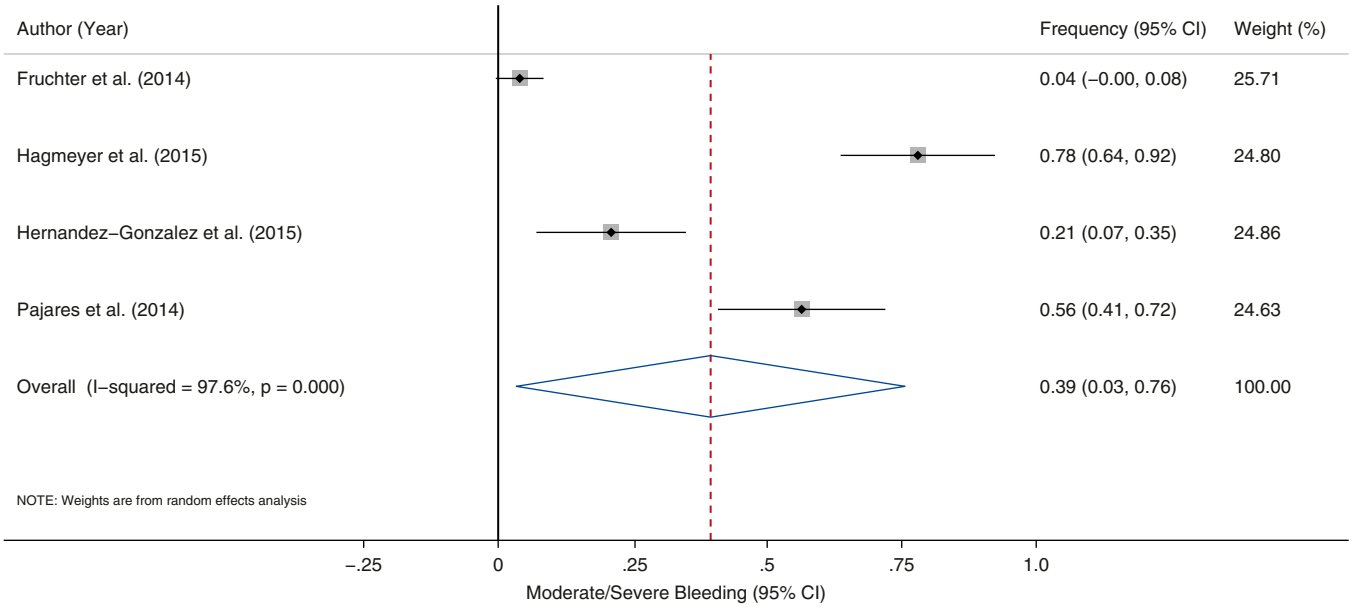

Figure 5. Forest plot of procedure-related moderate/severe bleeding. CI = confidence interval.

for hypersensitivity pneumonitis (32). Increasingly granular diagnostic modalities are needed to improve overall diagnostic accuracy, but they must be feasible, with an acceptable safety profile, if they are to become clinically useful. The role of transbronchial cryobiopsy in the diagnostic process may be as an intermediary step of evaluation, with surgical biopsy reserved for those in whom a confident diagnosis cannot be established after transbronchial cryobiopsy. Further characterization of the risks and benefits of transbronchial cryobiopsy, compared with currently used modalities, must be undertaken to define its role in the diagnostic evaluation of ILD.

Previous reviews have suggested that procedure-related complication rates are acceptably low with transbronchial cryobiopsy, being lower than those reported after surgical lung biopsy and comparable to those seen with forceps biopsy (8, 10). However, we identified significant heterogeneity across studies in the reported frequencies of pneumothorax (0–26%) and moderate/severe bleeding (0–78%). There are likely multiple reasons for this variability, including procedural technique (e.g., duration of freeze time, probe positioning), different approaches to post-biopsy monitoring (e.g., routine chest X-ray assessment for pneumothorax), inconsistent definitions of adverse events such as bleeding, and differences in study populations. Previous reports that examined the heterogeneity of complications in more detail have found that the risk of pneumothorax increases with the extent of lower lobe radiographic fibrosis, a histologic usual interstitial pneumonia pattern, and operator inexperience (8). Future studies are required to determine the complication rates in specific patient populations, the benefit of safety measures such as routine prophylactic placement of a Fogarty balloon or other endoscopic blocker, and the role of procedural training and accreditation programs. A comprehensive risk evaluation must be balanced against the diagnostic accuracy of transbronchial cryobiopsy, a characteristic that is currently unknown.

Limitations

To the best of our knowledge, our study is the most rigorous assessment of the role of transbronchial cryobiopsy, but there are some limitations to this metaanalysis. Although a comprehensive systematic literature review was performed, we cannot ensure that all relevant articles related to our study question were identified. We purposefully analyzed studies of higher quality to strengthen the robustness of our findings; however, this also resulted in the exclusion of conference abstracts from the metaanalysis, studies that were not specific for ILD, and studies with unclear outcome reporting. As a result, our findings relate to the use of transbronchial cryobiopsy in patients with suspected ILD and do not necessarily apply to other populations (e.g., immunocompromised patients, post–lung transplant patients). In addition, the significant heterogeneity across studies limits the confidence in pooled estimates for both diagnostic yield and complication rates.

Conclusions

Calls for the widespread adoption of transbronchial cryobiopsy in the evaluation of ILD should be tempered by the absence of studies reporting its diagnostic accuracy and the significant heterogeneity in both yield and complication rates. Surgical lung biopsy carries nonnegligible risk; however, these risks may be justifiable given the high diagnostic yield of the procedure and the subsequent impact on the diagnostic evaluation and management. Although the overall risks of transbronchial cryobiopsy appear to be relatively low, these must be balanced against its variable yield and complication rates. Importantly, the reported yield of cryobiopsy is not synonymous with diagnostic accuracy, and the ability of transbronchial cryobiopsy to identify a true underlying histologic pattern remains unknown.

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Future research is needed to establish the diagnostic accuracy of transbronchial cryobiopsy in the evaluation of ILD through head-to-head comparisons with the current gold standard of surgically obtained lung tissue reviewed in the context of a multidisciplinary discussion. Until the diagnostic accuracy of cryobiopsy is established more clearly, patients and practitioners will be challenged to incorporate risk-benefit analyses into their clinical decision making. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

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